

# Strategic coexistence theory for evolutionary game theory

Sang Woo Park<sup>1,\*</sup>

<sup>1</sup> School of Biological Sciences, Seoul National University, Seoul, Korea

\*Corresponding author: sangwoopark@snu.ac.kr

## Abstract

## Introduction

Understanding the mechanisms that promote the evolutionary stability of competing strategies is a fundamental aim in evolutionary biology. To this end, evolutionary game theory (EGT) and replicator dynamics have provided key foundations, allowing us to predict the outcome of strategic competition through payoff inequalities, invasion criteria, and stability analysis. However, less attention has been given to similarities between replicator dynamics and community ecology until recently (Tarnita and Traulsen, 2025), even though the classic replicator equation stems from the generalized Lotka-Volterra equation.

Interestingly, even a simple two-strategy game can yield heterogeneous outcomes that mirror outcomes in community ecology: the dominance of a singular strategy (prisoner’s dilemma and harmony games), coexistence (hawk–dove games), and bistability (stag-hunt games). Such differences in conditions that promote evolutionary stability and dominance of a singular strategy and those that permit strategy coexistence indicate that EGT can be understood from the perspective of coexistence theory. Recasting game dynamics as a coexistence problem immediately raises the following question: when two strategies coexist, does the corresponding mechanism stabilize or equalize competition? Even though EGT allows us to predict whether two strategies can coexist or not, a framework for quantifying coexistence mechanism between competing strategies remains elusive.

Modern coexistence theory (MCT) from community ecology presents a particularly useful framework for quantifying coexistence mechanisms. MCT predicts that coexistence of two competing entities, be they species or strategies, requires stabilizing niche differences to be greater than average fitness differences. In community ecology, stabilizing niche differences represent differences in factors that limit the growth of competing species, causing each species to limit itself more than it limits its competitor. In EGT, a strategy’s niche can be interpreted as strategic environments defined by the composition of competing strategies and the resulting interactions. Thus, stabilizing niche differences arise when each strategy is favored by a different strategic environment, such that the spread of each strategy creates conditions that limit its own relative growth and favor the recovery of its competitor. In contrast,

fitness differences represent differences in the ability of each strategy to compete under the limiting strategic environments created by its competitor. This differs from payoff differences in EGT, which depend on the underlying strategy frequency and therefore combine both stabilizing effects caused by changes in the strategic environment and residual fitness differences that favor one strategy over another.

So far, MCT has been primarily utilized for understanding species coexistence, but recent work showed that MCT can be extended beyond species competition, especially for studying pathogen strain competition, allowing one to directly quantify niche and fitness differences of competing entities, be they species or strains. Inspired by this effort, we present a Strategic Coexistence Theory (SCT) for predicting coexistence and exclusion of competing strategies in EGT by quantifying strategic niche and fitness differences. We begin by introducing SCT, which can be applied to any generic system of EGT described by replicator equations. We show that SCT can recover outcomes from classic, two strategy games, providing reinterpretation of classic results from coexistence-theoretic perspective. We then apply SCT to comparing five mechanisms for the evolution of cooperation, revealing that direct and indirect reciprocity destabilizes competition between cooperation and defection, thus pushing the system towards priority effect regime. Finally, we consider a case study of microbial cooperation, where nonlinear microbial growth can both stabilize and equalize competition between cooperation and defection, allowing coexistence of strategies.

## Strategic coexistence theory

We begin with a classic replicator equation, which allows us to model changes in the frequency of competing strategies based on their payoffs:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (1)$$

where  $x_i$  represents the frequency of strategy  $i$ ,  $\pi_i(\mathbf{x})$  represents the payoff of strategy  $i$  for a given strategy distribution  $\mathbf{x}$ , and  $\bar{\pi}(\mathbf{x}) = \sum_i \pi_i(\mathbf{x})x_i$  represents the average payoff. Since strategy frequencies must sum to 1 (i.e.,  $\sum_i x_i = 1$ ), a single-strategy equilibrium dominated by strategy  $i$  can be described by a unit vector  $\mathbf{e}_i$ , where all elements are equal to zero except for the  $i$ -th element. We use these single-strategy equilibria to define niche and fitness differences between two competing strategies  $i$  and  $j$ . However, because payoffs in many games can be negative, payoff ratios are difficult to interpret. Thus, we work on a positive multiplicative scale by exponentiating the payoff terms:

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x})). \quad (2)$$

For convenience, we refer to  $W_i(\mathbf{x})$  as the multiplicative payoff of strategy  $i$ .

73 First, we define the niche overlap  $\rho$  between strategies  $i$  and  $j$  as the ratio between  
 74 the geometric mean of each strategy's multiplicative payoff in the strategic environ-  
 75 ment that it creates when common,  $\sqrt{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}$ , and the geometric mean of  
 76 each strategy's multiplicative payoff in the environment generated by its competitor,  
 77  $\sqrt{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}$ . The corresponding niche difference is then given by

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (3)$$

78 When  $1 - \rho > 0$ , each strategy performs better, on average, in the environment  
 79 generated by its competitor than in the environment generated by itself. In other  
 80 words, intraspecific competition is stronger than interspecific competition.

81 Second, we define the fitness difference as the ratio between the geometric means  
 82 of the multiplicative payoff of each strategy across the two single-strategy environ-  
 83 ments:

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (4)$$

84 In other words, strategy  $j$  has an average competitive advantage over strategy  $i$   
 85 ( $\kappa_j/\kappa_i > 1$ ) when it performs better, on average, across the environments generated  
 86 by itself and its competitor. Then, analogous to MCT, mutual invasion requires the  
 87 fitness ratio to fall within the bounds set by niche overlap:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (5)$$

88 This decomposition allows us to understand how different mechanisms contribute to  
 89 the coexistence or exclusion of competing strategies.

## 90 Classic two-strategy games

91 As an illustrative example, we begin with classic two-strategy games, which can be  
 92 described by the following payoff matrix (Figure 1A):

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (6)$$

93 Here, the first and second rows represent strategies 1 and 2, respectively, and the  
 94 columns represent the strategy of the interacting opponent. Writing  $x_1$  to denote  
 95 the frequency of strategy 1, the expected payoffs are

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (7)$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (8)$$

Thus, the dynamics can be written as

$$\frac{dx_1}{dt} = x_1(1 - x_1) [(R - T)x_1 + (S - P)(1 - x_1)]. \quad (9)$$

The signs of  $T - R$  and  $S - P$  determine whether each strategy can invade the other when rare. Specifically, strategy 2 can invade strategy 1 when  $T > R$ , whereas strategy 1 can invade strategy 2 when  $S > P$ . These two invasion conditions partition the game space into four regimes (Figure 1B): prisoner's dilemma ( $T > R$ ,  $P > S$ ), hawk-dove game ( $T > R$ ,  $S > P$ ), stag-hunt game ( $R > T$ ,  $P > S$ ), and harmony game ( $R > T$ ,  $S > P$ ). In standard EGT, these outcomes are obtained by analyzing the stability of equilibrium conditions. For simple games, this analysis is straightforward.

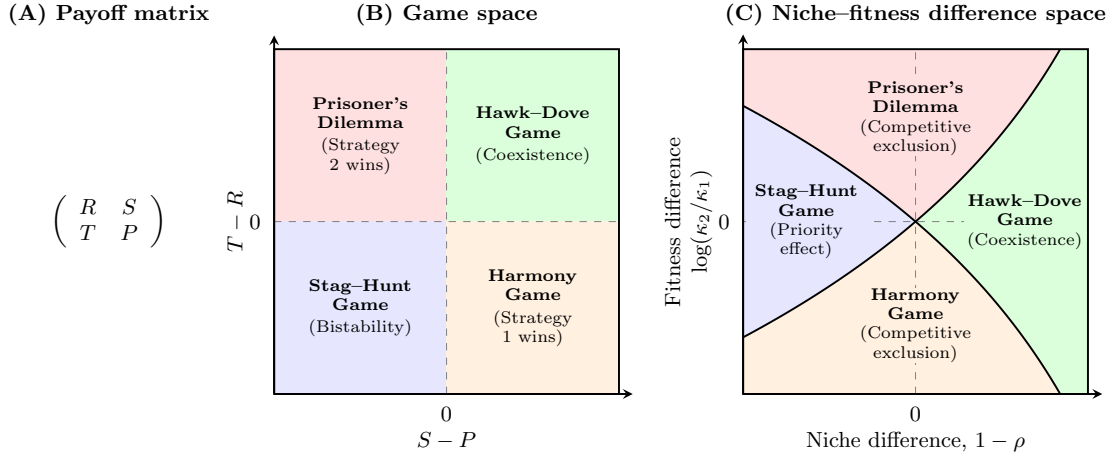


Figure 1: **Coexistence-theoretic interpretation of classic two-strategy games.** (A) General payoff matrix for a two-strategy game, where rows represent the focal strategy and columns represent the strategy of the interacting opponent. (B) Classification of four classic games based on the payoff differences  $T - R$  and  $S - P$ . Prisoner's dilemma and harmony games lead to the dominance of one strategy; hawk-dove games allow coexistence; and stag-hunt games result in bistability. (C) Mapping of the same four games onto niche-fitness difference space. Hawk-dove games correspond to the coexistence regime, where stabilizing niche differences overcome fitness differences. Prisoner's dilemma and harmony games correspond to competitive-exclusion regimes, whereas stag-hunt games correspond to the priority-effect regime.

A coexistence-theoretic approach recovers the same classification but provides a different interpretation (Figure 1C). For a two-strategy game determined by this

107 payoff matrix, the niche difference and fitness difference are given by

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2}\right), \quad (10)$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (11)$$

108 As shown above, mutual invasion requires  $\rho < \kappa_2/\kappa_1 < 1/\rho$ , which is equivalent to  
 109  $T > R$  and  $S > P$ . This corresponds to the hawk–dove game, in which each strategy  
 110 can invade when rare and the two strategies coexist. The remaining games also  
 111 have natural interpretations in niche–fitness difference space. Prisoner’s dilemma  
 112 and harmony games represent competitive exclusion regimes, where one strategy  
 113 has a fitness advantage that exceeds niche differences. In contrast, stag–hunt games  
 114 represent a priority-effect regime, where neither strategy can invade the other and  
 115 the outcome depends on initial conditions. Therefore, coexistence theory not only  
 116 allows us to recover the classic results from two-strategy games but also provides new  
 117 interpretations for these results in terms of niche and fitness differences. Building on  
 118 this, we next apply this framework to comparing core mechanisms of cooperation.

## 119 Comparisons of mechanisms for the evolution of co- 120 operation

121 Even though cooperative behavior can be found across a wide range of biological  
 122 systems, a simple prisoner’s dilemma shows that cooperation is not an evolutionarily  
 123 stable strategy. Thus, many researchers have sought to identify mechanisms that  
 124 promote evolutionary stability of cooperation. Here, we compare five classic mecha-  
 125 nisms for the evolution of cooperation using SCT and ask how they differ from the  
 126 perspective of strategy coexistence.

127 Specifically, we consider the following mechanisms: kin selection, direct reci-  
 128 procity, indirect reciprocity, network reciprocity, and group selection (Figure 2A).  
 129 Kin selection favors cooperation when two interacting individuals are genetic rela-  
 130 tives, where the genetic relatedness  $r$  must be greater than the cost-to-benefit ratio:  
 131  $r > c/b$ . Direct reciprocity favors cooperation when current cooperation may pro-  
 132 mote future cooperation by the same individual through repeated encounters. In  
 133 this case, the probability of repeated interaction (i.e., “next round”) must be greater  
 134 than the cost-to-benefit ratio:  $w > c/b$ . Indirect reciprocity favors cooperation when  
 135 current cooperation may promote future cooperation by different individuals through  
 136 reputation. Thus, the social acquaintanceship (i.e., the probability of knowing some-  
 137 one’s reputation) must be greater than the cost-to-benefit ratio:  $q > c/b$ . Network  
 138 reciprocity favors cooperation via clustering of cooperators, requiring the benefit-to-  
 139 cost ratio to be greater than the average number of neighbors:  $b/c > k$ . Finally,  
 140 group selection favors cooperation via multilevel selection, where groups with more  
 141 cooperators exhibit faster growth at the group level even if defectors reproduce faster

at the individual level. This requires  $b/c > 1 + (n/m)$ , where  $n$  is the maximum group size and  $m$  is the number of groups. All of these mechanisms can be described by a simple  $2 \times 2$  payoff matrix (Figure 2A), from which the corresponding niche and fitness differences can be computed using Eq. (10) and Eq. (11).

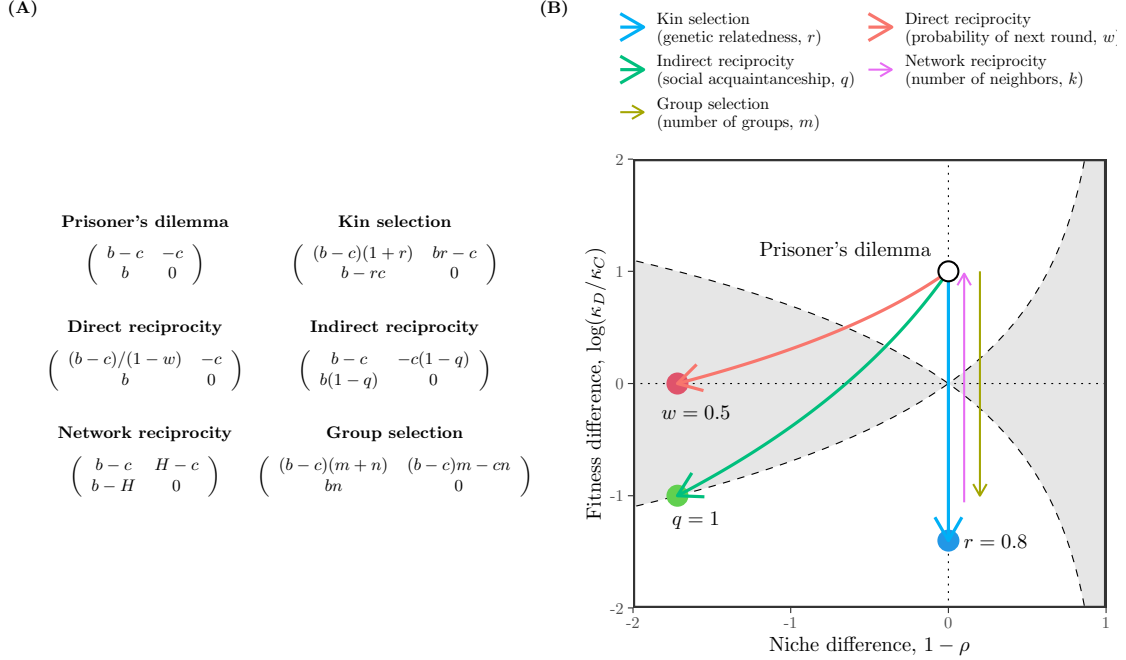


Figure 2: **Comparisons of five mechanisms for the evolution of cooperation using SCT.** (A) Payoff matrices for the baseline scenario (prisoner's dilemma) and five mechanisms for the evolution of cooperation (Nowak, 2006). Prisoner's dilemma is characterized by the benefit for the recipient  $b$  and the cost for the donor  $c$ . (B) Changes in niche and fitness differences driven by each mechanism. Each arrow represents the direction of change as we increase the corresponding parameter, indicated in figure legends. Note that kin selection, network reciprocity, and group selection all results in changes in fitness difference without causing any changes in niche difference. Therefore, we indicate the direction of change for network reciprocity and group selection separately to avoid overlapping arrows.

First, the prisoner's dilemma provides a baseline scenario, where defection is evolutionarily stable (Figure 2A). From the perspective of SCT, there is complete niche overlap, and therefore defection competitively excludes cooperation (Figure 2B). Building on this, SCT shows that five mechanisms promote evolutionary stability of cooperation via different dynamical routes (Figure 2B). For example, kin selection allows the fitness of cooperation to increase as genetic relatedness  $r$  increases (Figure 2B). However, it does not affect the niche difference between cooperation and defectors. Similarly, network reciprocity and group selection modulates the fitness

of cooperation without causing any changes in strategic niches. In contrast, direct and indirect reciprocity destabilizes competition while increasing the fitness of cooperation, thereby pushing the competition outcome towards a priority effect regime. Equivalently, these two mechanisms promote bistability of defection and cooperation.

SCT further reveals another difference between direct and indirect reciprocity. In particular, direct reciprocity is unable to generate sufficient fitness difference to competitively exclude defection even when the probability of next round  $w$  reaches 1. In contrast, indirect reciprocity is able to generate sufficient fitness difference to push the system towards the boundary between priority-effect regime and competitive-exclusion regime as social acquaintanceship  $q$  reaches 1, indicating that an additional mechanism will be able to promote complete dominance of cooperation.

We note that none of these mechanisms allow coexistence of cooperation and defection. Therefore, we move on to the final example, applying SCT to microbial cooperation, where nonlinear benefits and environmental context can promote coexistence of cooperators and defectors.

## Microbial cooperation and public goods

Here, we consider a simple, phenomenological model of competition between cooperating and cheating yeast in a cooperative sucrose-metabolism system, where invertase production by cooperators allows cheaters to utilize glucose, a public good released through sucrose hydrolysis (Gore et al., 2009). Specifically, invertase production by cooperators incurs a cost  $c$  and generates a total benefit of 1 that is captured privately with efficiency  $\epsilon$ . Assuming a linear relationship between glucose availability and microbial growth, the payoffs for cooperators  $\pi_C$  and cheaters (defectors)  $\pi_D$  can then be written as

$$\pi_C(f) = \epsilon + f(1 - \epsilon) - c, \quad (12)$$

$$\pi_D(f) = f(1 - \epsilon), \quad (13)$$

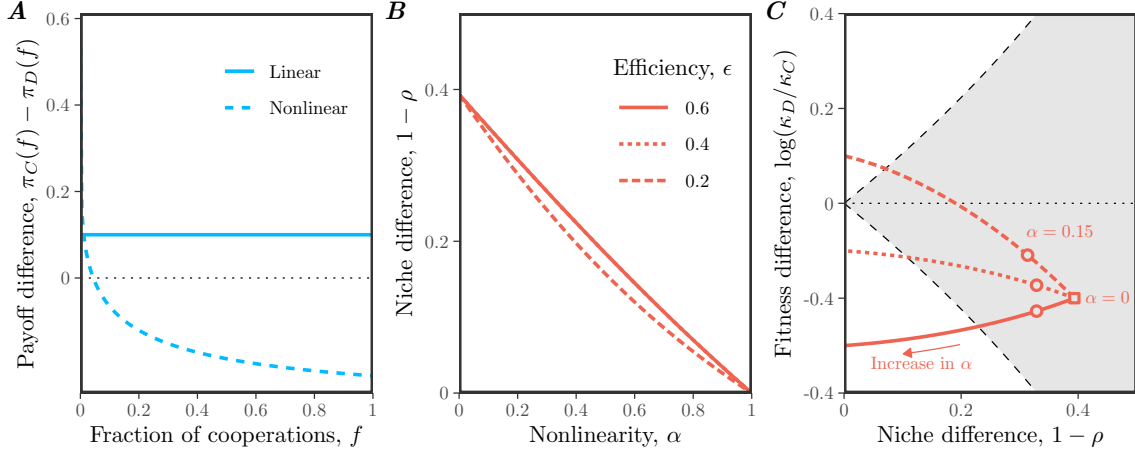
where the use of glucose depends on the fraction of cooperators  $f$ . The dynamics of cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\pi_C(f) - \pi_D(f)] \quad (14)$$

$$= f(1 - f)(\epsilon - c) \quad (15)$$

For this linear growth model, the payoff difference is fixed to  $\epsilon - c$  independent of  $f$  (Figure 2A). Therefore, cooperation is favored when efficiency is greater than the cost of cooperation,  $\epsilon > c$ , whereas defection is favored when the cost is greater than efficiency,  $c > \epsilon$  (Gore et al., 2009).

Our framework allows us to reinterpret this result from the perspective of coexistence theory. In particular, the two strategies have no niche difference, corresponding



**Figure 3: Effects of nonlinear microbial growth on coexistence of cooperation and defection.** (A) Payoff differences for linear and nonlinear models of microbial growth. In panel A, we assumed  $\epsilon = 0.4$ ,  $c = 0.3$ , and  $\alpha = 0.15$  for illustrative purposes. (B) Effect of nonlinearity  $\alpha$  on niche difference  $1 - \rho$ . Each line assumes a different value of efficiency  $\epsilon$ . Note that lines for  $\epsilon = 0.6$  and  $\epsilon = 0.4$  are completely overlapping. (C) Effect of nonlinearity  $\alpha$  on both niche difference  $1 - \rho$  and fitness difference  $\log(\kappa_D/\kappa_C)$ . The gray shaded area represents the coexistence regime. The square represents the condition when  $\alpha = 0$ . The circles represent the condition when  $\alpha = 0.15$ , which correspond to the experimental relationship between glucose and microbial growth (Gore et al., 2009). In panels B and C, we assumed  $c = 0.3$ , while allowing  $\epsilon$  and  $\alpha$  to vary.

186 to complete niche overlap,  $\rho = 1$ . In contrast, the fitness difference is determined by  
 187 the balance between private efficiency and the cost of cooperation:

$$\frac{\kappa_D}{\kappa_C} = \exp(c - \epsilon). \quad (16)$$

188 Therefore, this model always predicts competitive exclusion, except in the neutral  
 189 case  $c = \epsilon$ . The outcome is determined entirely by the fitness difference between  
 190 cooperators and defectors, rather than by stabilizing niche differences. This example  
 191 resembles the prisoner's dilemma discussed in the previous example (Figure 2).

192 In real biological systems, the effect of glucose on microbial growth can follow a  
 193 nonlinear relationship:

$$\pi_C(f) = [\epsilon + f(1 - \epsilon)]^\alpha - c, \quad (17)$$

$$\pi_D(f) = [f(1 - \epsilon)]^\alpha, \quad (18)$$

194 where  $\alpha = 0.15$  reflects the experimental relationship between glucose and microbial  
 195 growth (Gore et al., 2009). Then, the dynamics of cooperators and defectors are



described by the following replicator equation:

$$\frac{df}{dt} = f(1-f) [\{\epsilon + f(1-\epsilon)\}^\alpha - c - f(1-\epsilon)^\alpha]. \quad (19)$$

Under this nonlinear assumption, the payoff difference decreases with the cooperator fraction  $f$ , which can promote the coexistence of cooperators and defectors (Figure 2A): cooperators have a competitive advantage when rare, whereas defectors have a competitive advantage when cooperators are abundant.

From a coexistence-theoretic perspective, this means that nonlinear growth can generate sufficient stabilizing niche difference to overcome the fitness difference. In particular, we can show that the niche and fitness differences for this model are given by

$$1 - \rho = 1 - \exp\left(\frac{1 - \epsilon^\alpha - (1 - \epsilon)^\alpha}{2}\right), \quad (20)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(c + \frac{(1 - \epsilon)^\alpha - 1 - \epsilon^\alpha}{2}\right). \quad (21)$$

These expressions clarify the separate roles of cost, efficiency, and nonlinearity. For example, the cost of cooperation does not affect niche difference. Instead, niche difference is generated by the concave relationship between glucose availability and microbial growth, which increases from 0 to  $1 - \exp(-1/2)$  as  $\alpha$  goes from 1 to 0 (Figure 2B,C). In contrast, as  $\alpha \rightarrow 0$ , the fitness difference approaches

$$\log(\kappa_D/\kappa_C) \rightarrow c - \frac{1}{2}, \quad (22)$$

which no longer depends on the private capture efficiency  $\epsilon$  (Figure 2C). Thus, strong concavity not only generates stabilizing niche differences, but also acts as an equalizing mechanism (Figure 2C).

## Discussion

Standard evolutionary game theory and replicator equations provide powerful tools for predicting the outcome of strategy competition, including strategic dominance, coexistence, and bistability. However, these outcomes are typically understood by analyzing payoff differences within individual games, which makes it difficult to compare coexistence mechanisms across models. Building on Chesson's Modern Coexistence Theory from community ecology, we developed a strategic coexistence theory (SCT) that allows us to decompose strategy competition in terms of stabilizing niche differences and fitness differences. This decomposition provides a unifying framework for comparing different games from the perspective of coexistence theory, allowing us to identify stabilizing and equalizing mechanisms for strategy competition.

224 To demonstrate the utility of SCT, we considered three case studies. First, the  
225 analysis of classic two-strategy games showed that SCT recovers the original clas-  
226 sification for these classic games, highlighting the similarities between evolutionary  
227 games and community ecology. Then, we compared different mechanisms for the evo-  
228 lution of cooperation, which showed that each mechanism promotes cooperation via  
229 different dynamical routes. Kin selection, network reciprocity, and group selection  
230 primarily affects the fitness difference between cooperation and defection, whereas di-  
231 rect and indirect reciprocity can destabilize competition and generate priority effects.  
232 Finally, the analysis microbial cooperation revealed that

## 233 Supplementary Text

### 234 S1 Derivation of the strategic coexistence theory

235 To derive strategic coexistence theory (SCT), we begin with replicator equations:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (\text{S1})$$

236 where  $x_i$  represents the frequency of strategy  $i$ ,  $\pi_i$  represents the payoff of strategy  
 237  $i$ , and  $\bar{\pi}$  represents the average payoff. Then, the mutual invasion condition for two  
 238 competing strategies  $i$  and  $j$  can be written as

$$\pi_j(\mathbf{e}_j) < \pi_i(\mathbf{e}_j), \quad (\text{S2})$$

$$\pi_i(\mathbf{e}_i) < \pi_j(\mathbf{e}_i). \quad (\text{S3})$$

239 These inequalities will be preserved when we exponentiate the payoffs,  $W_i(\mathbf{x}) =$   
 240  $\exp(\pi_i(\mathbf{x}))$ :

$$W_j(\mathbf{e}_j) < W_i(\mathbf{e}_j), \quad (\text{S4})$$

$$W_i(\mathbf{e}_i) < W_j(\mathbf{e}_i). \quad (\text{S5})$$

241 Rearranging, we obtain the following inequality:

$$\frac{W_i(\mathbf{e}_i)}{W_j(\mathbf{e}_i)} < 1 < \frac{W_i(\mathbf{e}_j)}{W_j(\mathbf{e}_j)}. \quad (\text{S6})$$

242 By multiplying  $\sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}$  on both sides, we obtain:

$$\sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}} \quad (\text{S7})$$

243 Thus, by defining niche overlap  $\rho$  and fitness difference  $\kappa_j/\kappa_i$  as follows,

$$\rho = \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S8})$$

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S9})$$

244 we obtain the following inequality that defines conditions for mutual invasion and  
 245 long-term coexistence of two competing strategies:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S10})$$

## 246 S2 Applications to classic two strategy games

247 Consider a two strategy game described by the following payoff matrix:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (\text{S11})$$

248 There are four classic games that we can define from this simple payoff matrix:  
 249 prisoner's dilemma ( $T > R > P > S$ ), hawk-dove game ( $T > R, S > P$ ), stag-hunt  
 250 game ( $R > T, P > S$ ), and harmony game ( $R > T, S > P$ ). Here, we show that  
 251 each game corresponds to a distinct region in the niche-fitness difference space under  
 252 SCT.

253 To do so, we first begin by deriving expressions for niche overlap  $\rho$  and fitness  
 254 difference  $\kappa_j/\kappa_i$ . Writing  $x_1$  to denote the frequency of strategy 1, the expected  
 255 payoff for each strategy corresponds to:

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (\text{S12})$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (\text{S13})$$

256 Then, multiplicative payoffs for each strategy when rare can be written as:

$$W_1(1) = \exp(R) \quad (\text{S14})$$

$$W_1(0) = \exp(S) \quad (\text{S15})$$

$$W_2(1) = \exp(T) \quad (\text{S16})$$

$$W_2(0) = \exp(P) \quad (\text{S17})$$

257 Therefore, we have:

$$\rho = \exp\left(\frac{R + P - S - T}{2}\right), \quad (\text{S18})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (\text{S19})$$

258 As before, SCT allows us to predict the mutual invasion of two strategy using the  
 259 following inequality:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S20})$$

260 Then, simple algebra reveals that each condition in mutual invasion inequality  
 261 can be equivalently described by the payoff inequality:

$$\rho < \frac{\kappa_2}{\kappa_1} \iff R < T \quad (\text{S21})$$

$$\rho > \frac{\kappa_2}{\kappa_1} \iff R > T \quad (\text{S22})$$

$$\frac{\kappa_2}{\kappa_1} < \frac{1}{\rho} \iff P < S \quad (\text{S23})$$

$$\frac{\kappa_2}{\kappa_1} > \frac{1}{\rho} \iff P > S \quad (\text{S24})$$

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